

Anh (Joe) Nguyen

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SELECTED RESEARCH EXPERIENCE

Oregon State University — Research Asisstant

Jan 2022 - present

- LLM-based Computer use agent: Propose a simple yet effective method (RAG-PJ) to use multimodal tutorials for computer-use agents. We show that RAG-PJ achieves similar performance to other baselines *without using expensive world model rollouts*. [1]
- Language model-based RL agent: Developed an agent that can understand language descriptions of environment dynamics and act accordingly. The agent achieves *state-of-the-art policy generalization* over new language and unseen environment dynamics, increasing 6%-21% success rates over existing baselines ([Dynalang](#), [EMMA](#), [LWM](#)) [2]
- Visual language referring expression: Collected a new dataset in referring expressions for underwater objects. Compared and analyzed results from the state-of-the-art methods [OFA](#) and [CLIP](#): both have low accuracy performance on underwater objects that are rare in territorial settings.
- *Advised by Prof. [Stefan Lee](#)*

Singapore Management University — Research Engineer

Nov 2019-Nov 2021

- Developed a method that can clarify ambiguous language queries in visual domains (referring expression): detect the most ambiguous portion of the query based on neural module network, then raise an relevant clarifying question to users.
- The method results in more efficient clarifying process: less than 10% of clarifying questions than existing baselines in [CLEVR-Ref](#) and [Ref-Reasoning](#).
- *Advised by Prof [Jiang Jing](#)*

PUBLICATION

1. [Anh Nguyen](#), [Stefan Lee](#). [Look It Up Before Looking Ahead: Tutorials Reduce the Benefit of World Models in Computer-Use Agents](#), submitted to EMNLP 2026.
2. [Anh Nguyen](#), [Stefan Lee](#). [Language-conditioned world model improves policy generalization by reading environmental descriptions](#), accepted at Bridging Language, Agent, and World Models workshop at NeurIPS2025.
3. [Anh Nguyen](#), [Duy Tue Tran Van](#), [Minh Quoc Nghiem](#). [Bachelor Thesis: Extractive summarization with bidirectional encoder representations from transformers](#): achieves state-of-the-art *long-text* summarization on CNN/Daily Mail (2019).
4. [Duy Phung](#), [Tu Minh](#), [Anh Nguyen](#), [Tien Dinh](#), DTA Hunter System: A new statistic-based framework of predicting future demand for taxi drivers, accepted for presentation @ SoICT 2017 (The Eighth International Symposium on Information and Communication Technology)

EDUCATION

- 2022 - 2027 (Expect) Msc ([thesis](#)) and PhD (Artificial Intelligence) at **Oregon State University**
GPA: 3.9/4.0
Advised by Professor [Stefan Lee](#)
- 2014 - 2019 Bachelor's Degree at **VNU-Ho Chi Minh University of Science**
(GPA: 3.8/4.0, top 5% of Department)
Advised by Dr. [Nghiem Quoc Minh](#)

ACADEMIC PROJECTS

- **Language-conditioned LLM-based world model**: Identified the research gap of current LLM-based model based RL systems: they failed to incorporate language into the world model, thus unable to change world modeling on the fly based on language.
- **Exploration in RL**: Compared different exploration strategies in model-based RL: count-based, curiosity-based and Monte-Carlo dropout in PointMaze
- **Search in Games (Sudoku and (M,N,K))**: Compared different tree search in Sodoku and (M,N,K) game: Monte-Carlo Tree Search, A* and Minimax with Alpha-Beta pruning.

AWARDS

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| 2025 | Scholarly Presentation Award from Oregon State University. |
| 2022 | Vietnam Education Foundation (VEF) 2.0 candidate for Ph.D programs in the USA. |
| 2019 | Fully funded exchange at APEC Digital Innovation and Entrepreneurship Forum (Taiwan) |
| 2019 | Fully funded exchange at GKS ASEAN SCIENCE at Kyungwoon University (Korea) |
| 2019 | Fully funded exchange at KAIST (South Korea) |
| 2019 | Fully funded exchange at Southeast Asia Machine Learning School |
| 2019 | Scholarship from Shinhan Bank for excellent students |
| 2017 | Scholarship from American Chamber of Commerce Vietnam for excellent students |
| 2017 | Scholarship from Global Cybersoft for excellent students |
| 2017 | Scholarship from Esilicon Vietnam for excellent students |
| 2016 | ACM-ICPC Vietnam National Competitive programming: Third Prize |
| 2016 | HCMc University of Science Olympiad in Informatics : Honorable mention |
| 2014 | Viethope Scholarship for excellent college students |
| 2013 | National Vietnam Olympiad in Informatics: Honorable Mention |
| 2013 | Vallet scholarship for excellent students |

SKILLS

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| Programming | C++ (expert), Python (expert), Java (familiar with), competitive programming |
| Research (expert) | Pytorch, Jax, Tensorflow, deep learning, reinforcement learning, LLM |